STANDARD OPERATING PROCEDURE

Disk Error Diagnosis and Repair via Chkdsk

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1. Objective

To identify and repair filesystem structure corruption, bad logical storage blocks, and read/write drive anomalies causing performance bottlenecks.

2. Scope

Applies to client storage assets (HDDs and SSDs) experiencing extreme latency or error warnings.

3. Pre-requisites & Warnings

> **CRITICAL WARNING:** Running a deep check disk scan requires a volume lock, forcing a system reboot. The analysis can take 30 minutes to over 2 hours. Always secure user authorization regarding operational downtime before scheduling.

4. Step-by-Step Procedure

4.1 Open Command Prompt as Administrator

- Launch an elevated Command Prompt session via the Windows Start Menu search interface.

4.2 Execute a Safe Read-Only Disk Evaluation

- Perform a rapid, non-intrusive scan of the root target drive while the operating system is live:

```
chkdsk C:
```

- Review the terminal output summary to identify if logical structural faults exist.

4.3 Schedule Deep File System and Sector Repair

- If integrity or sector anomalies are found, run the deep correction sequence:

```
chkdsk C: /f /r
```

- *Parameter Key:* `/f` flags errors to be corrected; `/r` uncovers unreadable data from degraded physical blocks.

4.4 Authorize Volume Reboot Schedule

- When the console prompts: *"Would you like to schedule this volume to be checked the next time the system restarts? (Y/N)"*, type **Y** and hit **Enter**.

4.5 Force Reboot to Initialize Repair Scan

- Execute a direct reboot to pass control to the Windows boot scanner:

```
shutdown /r /t 0
```

- Remind the user explicitly not to intercept power connectivity while the verification percentages run.

5. Expected Outcome

Drive directory indices are corrected, corrupted disk structural pointers are patched, and bad storage blocks are securely quarantined out of service.

6. Escalation & Documentation

If the `chkdsk` loop routinely hangs indefinitely at a static progress percentage or trips hardware faults, immediately assign the case to the L2 Hardware Infrastructure Desk for drive replacement.